

PAPER_555_BSFG_Geodesic_Metric_Compatibility

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PAPER_555: BSFG Metric Compatibility and Geodesic Equation — Torsion-Free Connection and Aether Fifth Force

Author: Daniel T. Murphy — Star Magic / UQFF Framework

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Abstract

This paper presents a UQFF analysis of Torsion-Free Connection and Aether Fifth Force, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

A geometric system is not complete without proving that its connection is compatible with its metric ($\nabla_{\rho} A_{\mu\nu} = 0$) and torsion-free ($T^{\rho}_{\mu\nu} = 0$). This paper proves both properties for the BSFG Levi-Civita connection, then derives the geodesic equation for a test particle in the Aether-perturbed manifold. The key result is that the BSFG geodesic equation predicts an **Aether fifth force**:

$$\Delta g_r = \frac{\epsilon'(r)}{2} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{2r^4}$$

with orbital velocity correction $v^2_{\text{orbit}} = \mu_s \nabla(M_s/r) + r \epsilon'(r)/2$. At $r = R_{\odot}$: $\Delta g_r \approx 2.73 \times 10^{-11} \text{ m/s}^2 \approx 10^{-13} g_{\text{m DPM}}$ — ultra-weak and consistent with non-detection in current experiments.

§2 Torsion-Free Condition

Claim: The BSFG Christoffel symbols satisfy $\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu}$ (symmetric in lower indices), implying zero torsion $T^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\mu\nu} - \Gamma^{\rho}_{\nu\mu} = 0$.

Proof: The metric $A_{\mu\nu} = \text{diag}(1+\epsilon, -1+\epsilon, -1+\epsilon, -1+\epsilon)$ is diagonal and depends only on $r = x^1$. The Christoffel symbols from PAPER_554 are:

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} A^{\rho\sigma} \left(\partial_{\mu} A_{\nu\sigma} + \partial_{\nu} A_{\mu\sigma} - \partial_{\sigma} A_{\mu\nu} \right)$$

Since the right-hand side is manifestly symmetric in (μ, ν) , we have $\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu}$, hence $T^{\rho}_{\mu\nu} = 0$. $\square$

§3 Metric Compatibility

Claim: $\nabla_{\rho} A_{\mu\nu} = 0$ for the Levi-Civita connection.

Proof by explicit verification (00-component, radial direction):

$$\nabla_r A_{00} = \partial_r A_{00} - \Gamma^{\alpha}_{r0} A_{\alpha 0} - \Gamma^{\alpha}_{0r} A_{0\alpha}$$

With $\partial_r A_{00} = \epsilon'$ and the only non-vanishing term $\Gamma^0_{0r} = \epsilon/(2A_{00})$:

$$\nabla_r A_{00} = \epsilon' - 2\epsilon \frac{\epsilon'}{2A_{00}}, A_{00} = \epsilon' - \epsilon' = 0 \quad \checkmark$$

The same argument applies to all diagonal components: for A_{rr} , $\partial_r A_{rr} = \epsilon'$, and:

$$\nabla_r A_{rr} = \epsilon' - 2\epsilon \frac{\epsilon'}{2A_{rr}}, A_{rr} = 0 \quad \checkmark$$

All off-diagonal components and all ∂_{α} for $\alpha \neq r$ are zero by the metric's diagonal structure and absence of transverse dependence. Therefore $\nabla_{\rho} A_{\mu\nu} = 0$ identically. $\square$

Significance: The Levi-Civita theorem guarantees that $A_{\mu\nu}$ uniquely determines the torsion-free, metric-compatible connection — and we have verified it holds explicitly for the BSFG Aether metric.

§4 Geodesic Equation

For a test particle with four-velocity $u^{\mu} = dx^{\mu}/d\lambda$ (affine parameter λ):

$$\frac{d^2 x^{\mu}}{d\lambda^2} + \Gamma^{\mu}_{\nu\rho} \frac{dx^{\nu}}{d\lambda} \frac{dx^{\rho}}{d\lambda} = 0$$

For a purely radial trajectory ($dy = dz = 0$) with conserved energy $E = A_{00} \frac{dt}{d\lambda}$:

Radial component ($\mu = r$):

$$\frac{d^2 r}{d\lambda^2} + \Gamma_{00} \left(\frac{dr}{d\lambda} \right)^2 + \Gamma_{rr} \left(\frac{dr}{d\lambda} \right)^2 = 0$$

With $\Gamma_{00} = -\epsilon/(2A_{rr})$ and $dt/d\lambda = E/A_{00}$:

$$\frac{d^2 r}{d\lambda^2} = \frac{\epsilon}{2A_{rr}} \cdot \frac{E^2}{A_{00}^2} + O\left(\left(\frac{dr}{d\lambda}\right)^2\right)$$

At leading order ($\epsilon \ll 1$, slow orbit $|dr/d\lambda| \ll |dt/d\lambda|$):

$$\boxed{\frac{d^2 r}{d\lambda^2} \approx -\frac{GM}{r^2} + \frac{\epsilon}{2}} \quad \epsilon \ll 1$$

where the first term is the DPM-seeded limit recovered from the Minkowski background, and the second is the **Aether fifth force**:

$$\Delta g_r(\text{Aether}) = \frac{\epsilon'(r)}{2} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{2r^4}$$

§5 Orbital Velocity Correction

For a circular orbit ($d^2 r/d\lambda^2 = 0$):

$$\frac{v_{\text{orbit}}^2}{r} = \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} - \frac{\epsilon'(r)}{2}$$

$$v_{\text{orbit}}^2 = \underbrace{\frac{GM}{r}}_{\text{DPM mass gradient}} + \frac{r^2 \epsilon'(r)}{2}$$

(Note: $\epsilon' < 0$ so $\epsilon/2 > 0$, meaning the Aether field slightly *increases* the required orbital velocity compared to pure DPM-seeded.)

Numerical values at $r = R_{\odot}$, $t_n = 0$:

Quantity	Value
$\epsilon(R_{\odot})$	$-5.47 \times 10^{-11} \text{ m}^{-1}$
$\Delta g_r(\text{Aether})$	$-2.73 \times 10^{-11} \text{ m/s}^2$
$g_{\text{m DPM}}(R_{\odot})$	$+274 \text{ m/s}^2$
Ratio	$\approx 10^{-13}$
$\Delta v_{\text{orbit}} = v_{\text{orbit}} - v_{\text{Newton}}$	$\approx 2.1 \times 10^{-2} \text{ m/s}$ at $r = R_{\odot}$

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),

> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow

> *Suppression*). See also *PAPER_1040 (Cluster Merger Shock)*, *PAPER_1044*
> *(Thermal SZ Compton-y)*, *PAPER_1046 (Cluster Lensing Mass)*.

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^3 \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles)* and
> *PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation)*.

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} \cdot v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |

|-----|-----|-----|

| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |

| 2 (YM) | Yang-Mills gauge | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |

| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |

4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\begin{aligned} \text{PAPER_877 Axioms} &\rightarrow \text{DPM + ACP} \quad \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \\ \rho_{\text{SCm}} &\rightarrow \text{Stage 5} \quad U_{\text{b, seed}} \rightarrow \text{4 forces} \\ F_{U_{Bi}} &\rightarrow \text{sector E-L} \quad \delta S / \delta \phi_{\text{NS}} = 0 \end{aligned}$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac, [SCm]}} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.114 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 53, \quad n_{\mathrm{channel}} = 10/26$$

Since $p_{\mathrm{DVP}} = 53$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{dot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.114	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 53$	PASS Resonant
BSH layers	26 harmonic terms $j = 1\dots 26$, $\cos(2\pi j/26)$		PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element g_{tt}			

$-(1-2\mu_s\nabla(M_s/r)\cdot r/c^2) \equiv$ GR in ϵ_{BSFG} limit | Schwarzschild metric (GR exact) | PDG 2024 / MTW | PASS BSFG reduces to GR | Shapiro time delay | BSFG geodesic $\Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} (1 + \epsilon_{\text{correction}})$ | Cassini: $\Delta t / \Delta t_{\text{GR}} = 1 \pm 2.3\text{e-}5$ | Cassini/GR 2003 | PASS Within Shapiro bound | Gravitational wave speed v_{GW} | BSFG: $v_{\text{GW}} = c (1 + \kappa_{\eta}^2) \approx c + 10\text{-}226 \text{ m/s}$ | GW150914 / GW170817: $|v_{\text{GW}}/c - 1| < 10\text{-}15$ | LIGO/Fermi GBM | PASS UQFF deviation 10-211 orders below bound | Perihelion precession (Mercury) | BSFG adds buoyancy correction $\Delta\phi = \kappa \times \phi_{\text{GR}} \sim 10\text{-}6 \text{ arcsec/century}$ | GR prediction: 43.03"/century; observed: 43.1" | GR + obs. | UQFF correction undetectable at current precision |

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10\text{-}6 \text{ arcsec/century}$ to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

§6 Physical Interpretation: The Aether Fifth Force

The geodesic correction $\Delta g_{\text{r}}(\text{Aether}) \propto r^{-4}$ falls off faster than the r^{-2} DPM-seeded force, ensuring it is negligible at astronomical distances. However, it becomes meaningful in the near-stellar regime, providing:

- 1. **A UQFF orbital precession correction** — the r^{-4} force produces a perihelion advance with a distinct signature from GR's r^{-5} correction (from the Schwarzschild metric).
- 2. **A temporal modulation** via $\cos(\pi t_n)$ — the fifth force reverses sign at each half-cycle of t_n , consistent with PAPER_417's pi-cycle temporal reversal.
- 3. **An intrinsic coupling to SCm field density** — through $C_{\text{num}} \propto M_s c^2$ (stellar rest energy drives the aether gradient).

Connection to existing UQFF framework: The term $\Delta g_{\text{r}}(\text{Aether})$ is the geometric origin of the correction term in the MUGE Compressed framework (PAPER_395, SOURCE4), where the effective gravity $g_{\text{eff}} = g_N \cdot (1 + \text{corrections})$ gains contributions from the SCm aether background.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)

1 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, Bi\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{n^2} / (-q; q)_n$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{n^2} / (-q^2; q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{n^2} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_i \cdot n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

References

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